

I. K. Gujral Punjab Technical University
Bachelor of Computer Applications (BCA)

2.	Create a warehouse application using tools supplied by any SaaS provider.
3.	Implementation of Para-Virtualization using VM Ware's Workstation/ Oracle's Virtual Box and Guest O.S. Learn creation, migration, cloning and managing of virtual machines.
4.	Using public cloud service providers tools for exploring the usage of IaaS, PaaS and SaaS cloud services.
5.	Setting up a private cloud using open source tools (Eucalyptus/Open Stack etc.).

Course Code: UGCA1943

Course Name: Android Programming

Program: BCA	L: 3 T: 0 P: 0
Branch: Computer Applications	Credits: 3
Semester: 6 th	Contact hours: 33 hours
Theory/Practical: Theory	Percentage of numerical/design problems: --
Internal max. marks: 40	Duration of end semester exam (ESE): 3hrs
External max. marks: 60	Elective status: Skill Enhancement
Total marks: 100	

Prerequisite: The course will help students to develop applications for Android Mobile Phone. The students will use a software emulator for the phone to develop the application and a real phone to demonstrate the application. The main emphasis is on the aspects like develop, debug and test a variation of an existing application. Students must know all the basic concepts of Java.

Co requisite: -NA-

Additional material required in ESE: Students can carry their own data cable to execute the application built on Simulator for the sake of fast speed.

Course Outcomes:

CO#	Course outcomes
CO1	Prepare environment for working on Android OS.
CO2	Highlight various security issues in Android platform.
CO3	Design innovative User Interface and develop activity for android app.
CO4	Outline the steps for creating database applications.
CO5	Write programs for basic Android based applications.

Detailed contents	Contact hours
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Unit-I Characteristics of Mobile applications, Introduction to Android Development Environment, Advantages and Futures of Android, Architecture and working of Android, User-interface design for mobile applications and managing application data. [CO1] [CO3]	8
Unit-II Integrating cloud services, networking, OS and hardware into mobile-applications. Enterprise requirements in mobile applications: Performance, Scalability, Modifiability, Availability and Security. [CO2]	7
Unit-III Mobile Software Engineering (Design Principles, Development, Testing methodologies for mobile applications).[CO4]	7
Unit-IV Directory Structure of an Android Project, Common Default Resources Folders, The Values Folder, Leveraging Android XML.[CO5]	8

Text Books:

1. Android Studio Application Development, Belen Cruz, Zapata, Packt Publishing
2. Deitel, P., Deitel, H., Deitle, A., and Morgano, M., Android for Programmers – An App-Driven Approach, Prentice Hall

Reference Books:

1. Professional Mobile Application Development, JEFFMCWHERTER, SCOTTGOWELL, Wiley.
2. Professional Android 4 Application Development, Reto Meier, Wrox Publication
3. Beginning iPhone Development with Swift, David Mark, A press Publication

E Books/ Online learning material

1. d.android.com
 2. Safari Textbooks Online: <http://library.ohio-state.edu/search/y?SEARCH=Safari>
 3. <https://www.androidauthority.com/best-ebook-ereader-apps-for-android-170696/>
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